

Day 3 — Wazuh SIEM/XDR Deployment

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Objective

To deploy a fully functional Wazuh SIEM/XDR platform (Manager, Indexer, and Dashboard), register agents across multiple operating systems, and validate end-to-end threat detection by triggering a real brute force attack and confirming correct alert generation with MITRE ATT&CK mapping.

System Information

Detail	Value
Wazuh Server Host	Ubuntu Server 22.04 LTS — 192.168.56.104
Deployment Method	Docker (all-in-one, single-node)
Wazuh Version	4.9.0
Agents Registered	3 (Kali, Windows 10, Ubuntu Server)

Practical Work Completed

Deployment workflow

Docker was installed on Ubuntu Server, followed by cloning Wazuh's official Docker repository (v4.9.0) and generating SSL certificates for secure inter-component communication. The full Wazuh stack — Manager, Indexer, and Dashboard — was then deployed as three Docker containers using a single `docker compose up -d` command.

1. Downloaded the Docker installation script, Installed and started Docker Engine.

```
Session Actions Edit View Help
fara@ubuntuserver:~$ curl -fsSL https://get.docker.com -o get-docker.sh
sudo sh get-docker.sh

+ sh -c systemctl enable --now docker.service 2>/dev/null
INFO: Docker daemon enabled and started

+ sh -c docker version
Client: Docker Engine - Community
Version: 29.6.1
API version: 1.55
Go version: go1.26.4
Git commit: 8900f1d
Built: Fri Jun 26 11:40:26 2026
OS/Arch: linux/amd64
Context: default

Server: Docker Engine - Community
Engine:
Version: 29.6.1
API version: 1.55 (minimum version 1.40)
Go version: go1.26.4
Git commit: 8ec5ab3
Built: Fri Jun 26 11:40:26 2026
OS/Arch: linux/amd64
Experimental: false
containerd:
Version: v2.2.5
GitCommit: e53c7c1516c3b2bff98eb76f1f4117477e6f4e66
runc:
Version: 1.3.6
GitCommit: v1.3.6-0-g491b69ba
docker-init:
Version: 0.19.0
GitCommit: de40ad0
```

2. Cloned the Wazuh Docker repository (v4.9.0) from GitHub.

```
fara@ubuntuserver:~$ git clone https://github.com/wazuh/wazuh
-docker.git -b v4.9.0
Cloning into 'wazuh-docker' ...
remote: Enumerating objects: 18545, done.
remote: Counting objects: 100% (66/66), done.
remote: Compressing objects: 100% (23/23), done.
remote: Total 18545 (delta 50), reused 49 (delta 43), pack-re
used 18479 (from 2)
Receiving objects: 100% (18545/18545), 7.71 MiB | 2.04 MiB/s,
done.
Resolving deltas: 100% (10306/10306), done.
Note: switching to 'cb63566719ce2e78b9d3c111a0a61d743fc699fc'
•
You are in 'detached HEAD' state. You can look around, make e
```

3.Added the current user (fara) to the Docker group.

```
fara@ubuntu-server:~$ sudo usermod -aG docker fara
fara@ubuntu-server:~$
```

4.Navigated to the single-node deployment directory.

```
fara@ubuntu-server:~$ ls
get-docker.sh  wazuh-docker
fara@ubuntu-server:~$ cd wazuh-docker/
fara@ubuntu-server:~/wazuh-docker$ ls
build-docker-images  LICENSE  SECURITY.md
CHANGELOG.md         multi-node  single-node
indexer-certs-creator  README.md  VERSION
fara@ubuntu-server:~/wazuh-docker$ cd single-node/
fara@ubuntu-server:~/wazuh-docker/single-node$ ls
config  generate-indexer-certs.yml
docker-compose.yml  README.md
```

5.Generated SSL/TLS certificates for Wazuh components.

```
Session Actions Edit View Help
fara@ubuntu-server:~/wazuh-docker/single-node$ docker compose
-f generate-indexer-certs.yml run --rm generator
WARN[0000] /home/fara/wazuh-docker/single-node/generate-indexer-certs.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] 1/1te 1/1
✓ Network single-node_default Created 0.1s
Image wazuh/wazuh-certs-generator:0.0.2 Pulling
```

```
7ce91ec7d1d3 Pulling ID
5249716d429c Pull complete 0B
7ce91ec7d1d3 Pull complete 0B
d7003467fd14 Pull complete 0B
Image wazuh/wazuh-certs-generator:0.0.2 Pulled
Container single-node-generator-run-9016afef1769 Creating
Container single-node-generator-run-9016afef1769 Created
The tool to create the certificates exists in the in Packages
bucket
01/07/2026 13:00:14 INFO: Generating the root certificate.
01/07/2026 13:00:14 INFO: Generating Admin certificates.
01/07/2026 13:00:14 INFO: Admin certificates created.
01/07/2026 13:00:14 INFO: Generating Wazuh indexer certificates.
01/07/2026 13:00:14 INFO: Wazuh indexer certificates created.
01/07/2026 13:00:14 INFO: Generating Filebeat certificates.
01/07/2026 13:00:14 INFO: Wazuh Filebeat certificates created
.
01/07/2026 13:00:14 INFO: Generating Wazuh dashboard certificates.
01/07/2026 13:00:14 INFO: Wazuh dashboard certificates created.
Moving created certificates to the destination directory
Changing certificate permissions
Setting UID indexer and dashboard
Setting UID for wazuh manager and worker
```

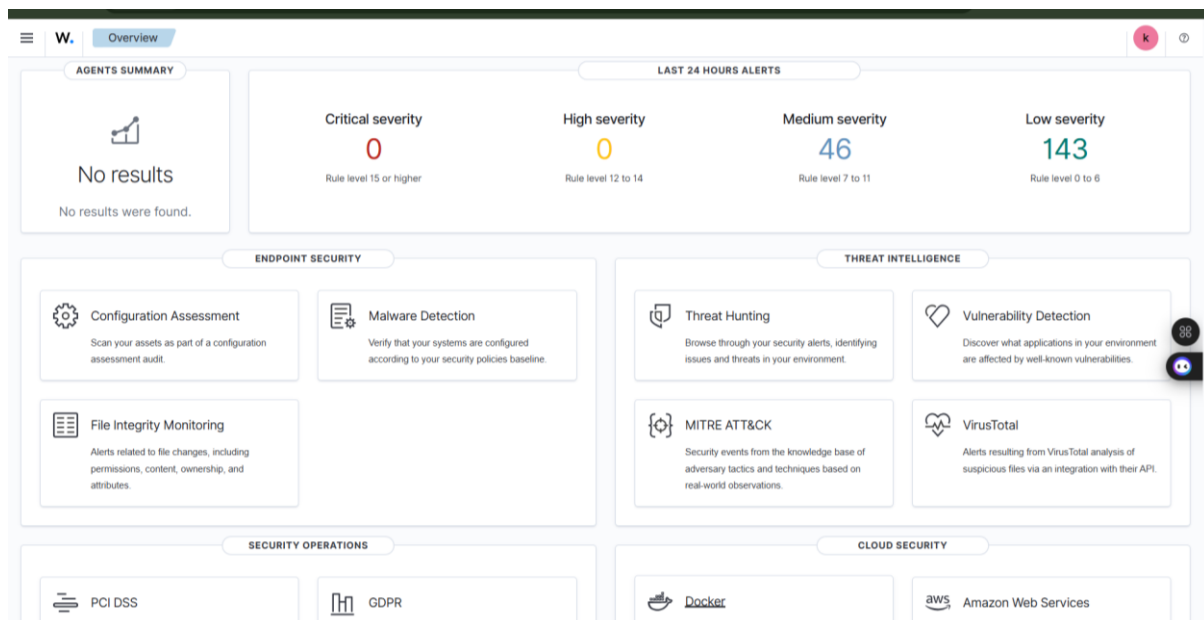
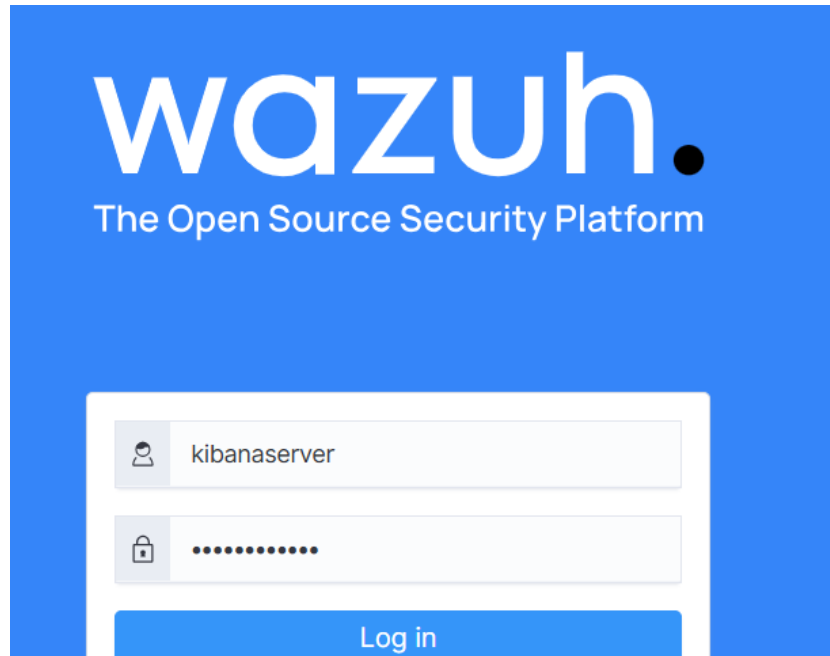
6.Started the Wazuh Manager, Indexer, and Dashboard using Docker Compose and Verified that all Docker containers were running successfully.

```
fara@ubuntu:~/wazuh-docker/single-node$ docker compose up -d
WARN[0000] /home/fara/wazuh-docker/single-node/docker-compose.yml: the
attribute `version` is obsolete, it will be ignored, please remove it t
o avoid potential confusion
[+] up 62/62
✓ Image wazuh/wazuh-indexer:4.9.0 Pulled 2827.3s
✓ Image wazuh/wazuh-dashboard:4.9.0 Pulled 123.4s
✓ Image wazuh/wazuh-manager:4.9.0 Pulled 678.0s
✓ Network single-node_default Created 0.1s
✓ Volume single-node_wazuh_api_configuration Created 0.0s
✓ Volume single-node_wazuh_wodles Created 0.0s
✓ Volume single-node_filebeat_var Created 0.0s
✓ Volume single-node_wazuh-indexer-data Created 0.1s
✓ Volume single-node_filebeat_etc Created 0.0s
✓ Volume single-node_wazuh-dashboard-custom Created 0.0s
✓ Volume single-node_wazuh_agentless Created 0.0s
✓ Volume single-node_wazuh_logs Created 0.0s
✓ Volume single-node_wazuh_integrations Created 0.0s
✓ Volume single-node_wazuh-dashboard-config Created 0.0s
✓ Volume single-node_wazuh_active_response Created 0.0s
✓ Volume single-node_wazuh_etc Created 0.0s
✓ Volume single-node_wazuh_queue Created 0.0s
✓ Volume single-node_wazuh_var_multigroups Created 0.0s
✓ Container single-node-wazuh.indexer-1 Started 13.4s
✓ Container single-node-wazuh.manager-1 Started 13.2s
✓ Container single-node-wazuh.dashboard-1 Started 2.2s
fara@ubuntu:~/wazuh-docker/single-node$
```

```
Session Actions Edit View Help
fara@ubuntu:~/wazuh-docker/single-node$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
NAMES
cf2a5f495438   wazuh/wazuh-dashboard:4.9.0        "/entrypoint.sh"        2 hours ago   Up 2 hours   443/tcp, 0.0.0
.0:443→5601/tcp, [::]:443→5601/tcp
single-node-wazuh.dashboard-1
c9ffd5e4ba4e   wazuh/wazuh-indexer:4.9.0          "/entrypoint.sh open..." 2 hours ago   Up 2 hours   0.0.0.0:9200→
9200/tcp, [::]:9200→9200/tcp
single-node-wazuh.indexer-1
427746d05000   wazuh/wazuh-manager:4.9.0          "/init"                  2 hours ago   Up 2 hours   0.0.0.0:1514-1
515→1514-1515/tcp, [::]:1514-1515→1514-1515/tcp, 0.0.0.0:514→514/udp, [::]:514→514/udp, 0.0.0.0:55000→5500
0/tcp, [::]:55000→55000/tcp, 1516/tcp
single-node-wazuh.manager-1
fara@ubuntu:~/wazuh-docker/single-node$
```

Dashboard Access

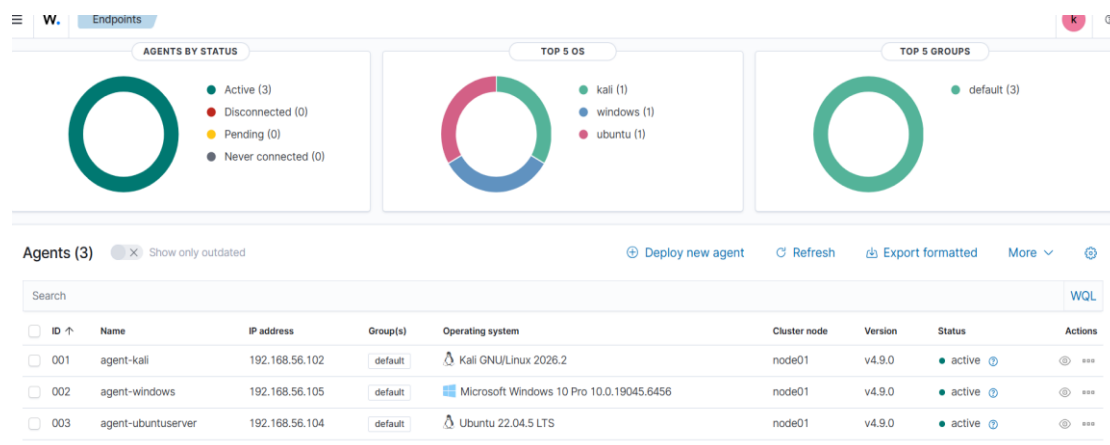
The Wazuh Dashboard was accessed via HTTPS at <https://192.168.56.104> and confirmed live and operational.



Agent Registration

Three Wazuh agents were deployed and registered:

- **agent-kali** — Linux agent on Kali VM
- **agent-windows** — Windows agent on Windows 10 VM
- **agent-ubuntuserver** — Linux agent on the Wazuh Manager host itself, demonstrating self-monitoring capability



Brute Force Detection Test

Attack Simulation

Five failed SSH login attempts were executed from Kali against Ubuntu Server (192.168.56.104) using incorrect passwords, simulating a real-world credential brute force attack.

- `ssh fara@192.168.56.104`

```
(fara@kali) - [~/Desktop]
$ ssh fara@192.168.56.104
Ubuntu 22.04.5 LTS
fara@192.168.56.104's password:
Permission denied, please try again.
fara@192.168.56.104's password:
Permission denied, please try again.
fara@192.168.56.104's password:
fara@192.168.56.104: Permission denied (publickey,password).

(fara@kali) - [~/Desktop]
$ ssh fara@192.168.56.104
Ubuntu 22.04.5 LTS
fara@192.168.56.104's password:
Permission denied, please try again.
fara@192.168.56.104's password:
Permission denied, please try again.
fara@192.168.56.104's password:
fara@192.168.56.104's password: 
```

Initial Issue and Resolution

No alerts initially appeared in the dashboard despite the failed logins being correctly recorded in Ubuntu Server's authentication log. Investigation confirmed all agents were active and the Manager's detection rules were functioning correctly when tested manually. The root cause was identified in the Ubuntu Server agent's configuration file (`ossec.conf`): the agent was not configured to monitor `/var/log/auth.log`, the file where SSH login failures are recorded. As a result, this data was never sent to the Manager, so no alert could be generated regardless of the rules in place.

Fix applied — the following block was added to the agent configuration:

```
<localfile>
  <log_format>syslog</log_format>
  <location>/var/log/auth.log</location>
</localfile>
```

The agent was restarted, after which alerts began generating correctly and immediately.

Verified Alert Output

Following the fix, the brute force attempt was successfully detected and logged with the following confirmed detections:

Rule ID	Description
5760	sshd: authentication failed

MITRE ATT&CK Mapping Confirmed:

- **T1110.001** — Brute Force: Password Guessing (Tactic: Credential Access)
- **T1021.004** — Remote Services: SSH (Tactic: Lateral Movement)

rule.id:5760							
manager.name: wazuh.manager agent.id: 003 + Add filter							
8 hits							
Jul 2, 2026 @ 00:50:35.325 - Jul 2, 2026 @ 01:00:35.325							
Export Formatted 488 columns hidden Density 1 fields sorted Full screen							
	timestamp	rule.mitre.id	rule.mitre.tactic	rule.description	rule.level	rule.id	
	Jul 2, 2026 @ 00:55:45.722	T1110.001 T1021.004	Credential AccessLateral Movem...	sshd: authentication failed.	5	5760	
	Jul 2, 2026 @ 00:55:45.722	T1110.001 T1021.004	Credential AccessLateral Movem...	sshd: authentication failed.	5	5760	
	Jul 2, 2026 @ 00:55:43.719	T1110.001 T1021.004	Credential AccessLateral Movem...	sshd: authentication failed.	5	5760	
	Jul 2, 2026 @ 00:54:27.448	T1110.001 T1021.004	Credential AccessLateral Movem...	sshd: authentication failed.	5	5760	
	Jul 2, 2026 @ 00:53:17.209	T1110.001 T1021.004	Credential AccessLateral Movem...	sshd: authentication failed.	5	5760	
	Jul 2, 2026 @ 00:52:53.161	T1110.001 T1021.004	Credential AccessLateral Movem...	sshd: authentication failed.	5	5760	
	Jul 2, 2026 @ 00:52:47.150	T1110.001 T1021.004	Credential AccessLateral Movem...	sshd: authentication failed.	5	5760	
	Jul 2, 2026 @ 00:52:35.152	T1110.001 T1021.004	Credential AccessLateral Movem...	sshd: authentication failed.	5	5760	

Wazuh Architecture — In My Own Words

Wazuh is an open-source SIEM and XDR platform built from three core components working together. The Wazuh Manager is the central brain — it receives all incoming security data from agents, applies detection rules against that data, and generates alerts when something suspicious is found. In our deployment, the Manager runs on Ubuntu Server.

The Wazuh Indexer is responsible for storing and indexing all the alerts and events the Manager generates, making them fast to search and retrieve later.

The Wazuh Dashboard is the visual interface analysts use to view everything — security events, agent status, integrity monitoring results, and MITRE ATT&CK mappings — all presented through graphs, tables, and searchable views.

On every machine we want to monitor, we install a lightweight Wazuh Agent. The agent collects logs, monitors file changes, and watches system activity locally, then forwards everything back to the Manager over a secure connection on ports 1514-1515.

In our lab, we deployed all three server-side components together using Docker, which packaged the Manager, Indexer, and Dashboard into containers started with a single command rather than installing each piece manually. We then registered three agents — one on Kali Linux, one on Windows 10, and one on Ubuntu Server itself — demonstrating that a Wazuh deployment can monitor both remote endpoints and its own host simultaneously.

Conclusion

Day 3 successfully established a fully operational Wazuh SIEM/XDR deployment using Docker, with three active agents monitoring Linux and Windows endpoints as well as the Manager host itself. A real brute force attack was simulated and, after diagnosing and resolving a log source misconfiguration on the Ubuntu Server agent, was correctly detected and mapped to MITRE ATT&CK technique T1110.001 (Brute Force) with the appropriate Credential Access tactic. This exercise demonstrated not only successful SIEM deployment but practical detection engineering troubleshooting — verifying each stage of the log pipeline from source to dashboard to identify and resolve a real configuration gap, a core skill for SOC and detection engineering work.